Sector A branch-aware T6 conditional-composition theorem

TECT verification-first repository · claim A5-SECTOR-A-SYNTHESIS

first issued 2026-07-19 · this version issued 2026-07-20 · v1.1

1. Purpose and scope: enacted conditional theorem

This document is the operator-confirmed issue of the Sector-A conditional- composition theorem. Jusang Lee confirmed the exact v1.0 source and PDF on 2026-07-20 and authorized creation of the PUBLISHED T6 bundle. The confirmation changes the governance state; it does not enlarge the mathematical statement.

Theorem A5-T6. On the fixed spectral three-torus with periods 16, assume the seven named hypotheses in Section 3 and the six content-addressed premises in Section 4. Then the following two branch-specific statements hold simultaneously.

1. The canonical three-complex-component production functional generates the unique global H^2 gradient flow supplied by A2, with its exact energy law, continuous dependence, and positive-time smoothing. On the declared positive-time solution balls, the restarted exact-Galerkin trajectories of full-production A3 converge with the residual and trajectory bounds proved in that premise.
2. On the distinct positive real-scalar branch, scalar A3 gives spectral cutoff removal at every fixed perturbative order and external momentum, while A4 gives full-sequence finite-volume sharp-spectral Gibbs convergence, lifted L^1/total-variation convergence, and convergence of the declared smeared cylinder-polynomial correlations.

The first statement is an implication chain. The second is a conjunction of independent theorems. Neither shared geometry nor a scalar reduction identifies the shell masses, state spaces, functionals, or measures of the two branches. This theorem is T6 because its exact premises and named hypotheses are exposed, the logical composition is independently audited, and the exact referee package has been confirmed. It is not a new full Class-II constructive measure.

1 Immutable T5 history and confirmed identity

The T6 theorem is published beside, not over, the immutable T5 capstone

```
claims/A5-SECTOR-A-SYNTHESIS/bundle/
  A5-Sector-A-Synthesis-T5-260719
```

The T5 bundle has 155 hash-listed files and content digest

```
5cf4397c38fb316ec108447404531e649e628d6fcc62d67e613d060b70b24ea5
```

Both T6 audits reconstruct every T5 file hash and this digest. Any drift falsifies the T6 composition instead of rewriting tier history.

The canonical JSON theorem contract has SHA-256

```
df01a1a3606d979307ac0bb8c9de14a4ab2d68fd83d228ed38f9e470eba823fc
```

The confirmation record additionally pins candidate commit

`fb776bff6b161178a6328570af3ef9529b44a2df`

and the exact v1.0 source and PDF hashes. Thus a later reissue cannot be mistaken for the artifact the operator reviewed.

2 Seven named hypotheses

The exact hypothesis set is:

1. `A5-H1-CANONICAL-KERNEL-MANIFEST`: the T5 kernel manifest is the canonical scalar spectral anchor.
2. `A1-KERNEL-CONV`: the Fourier and kernel conventions are fixed.
3. `A1-SHELL-POSITIVITY`: the scalar shell mass is positive.
4. `A2-H2-SEXTIC-COERCIVITY`: the positive sextic term controls the declared negative quartic term.
5. `A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL`: the exact T5 full variational realization is canonical for A2 and full A3.
6. `A3-H1-DIM3-Q4-KERNEL`: the scalar covariance has the declared three-dimensional fourth-order ultraviolet decay.
7. `A3-H2-IR-POSITIVITY`: the scalar covariance is infrared positive.

Items 1 and 5 are explicit T6 lifts for the two T5 hard premises. They do not relabel the underlying premise cards.

3 Six exact premises and branch topology

Record	Tier	Role
`A1-PRODUCTION-KERNEL-MANIFEST`	T5	scalar anchor through A5-H1
`A1-PRODUCTION-FUNCTIONAL-REALISATION`	T5	full functional through A2-H3
`A2-FULL-PRODUCTION-WELLPOSED`	T6	global full gradient flow
`A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM`	T6	positive-time Galerkin limit
`A3-PERTURBATIVE-CONTINUUM-CORRELATORS`	T6	fixed-order scalar cutoff removal
`A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE`	T6	finite-volume scalar Gibbs limit

Every component card, terminal manifest, and PUBLISHED support bundle is rehash-checked. The full branch has the directed chain

$$F_{\text{canonical}} \Longrightarrow \text{A2 flow} \Longrightarrow \text{A3 exact-Galerkin approximation.} \tag{4.1}$$

The scalar branch has only the conjunction

$$\text{fixed-order perturbative convergence} \quad \wedge \quad \text{finite-volume constructive convergence.} \tag{4.2}$$

There is no cross-proof arrow between the two terms of (4.2), and neither branch supplies a theorem about the other branch's field space.

4 Composition proof and scope controls

The three manifests agree on the declared torus, Fourier convention, q_0, Y, and $\eta_{\text{shell}} = 0$ interface, with the stored Z values agreeing within the declared 10^{-9} kernel tolerance. This proves compatibility of the statements on one geometry, not parameter identity.

By the canonical-functional hypothesis, A2 applies to the precise functional used by full A3. Uniqueness identifies A3's continuum trajectory with the A2 solution. Applying A3 on each declared $0 < \tau \leq t \leq T$ solution ball gives (4.1), including its residual and finite-time trajectory controls. It does not give a $t = 0$ algebraic rate for merely H^2 data or practical error bars for historical N32/N64/N128 solver outputs.

Under the kernel, positivity, coercivity, and ultraviolet hypotheses, scalar A3 and A4 each apply in their own exact scopes. Their two conclusions can therefore be asserted jointly as (4.2), but perturbative cutoff removal does not prove resummation or construct A4. A4 remains finite-volume and covers smeared cylinder-polynomial observables, not unsmeared composite fields.

Finally, the pinned scalar kernel gives

$$m_{\text{sc}}^2 = 0.005,$$

whereas the full-production manifest gives

$$m_{\text{full}}^2 = r - \frac{Z^2}{4Y} = 0.260000000009475.$$

Independent Decimal reconstruction enforces

$$|m_{\text{full}}^2 - m_{\text{sc}}^2| = 0.255000000009475 > 0.2. \tag{5.1}$$

The local scalar-reduction relative error is below 10^{-12} when the non-scalar structures are switched off. This checks the local convention; it does not define derivative-current products on the A4 Gaussian support or construct a three-component Class-II Gibbs measure. Equations (4.1), (4.2), and (5.1) prove exactly Theorem A5-T6 and its non-implications. □

5 Weakness map and highest-value open target

The package controls T5 premise visibility, immutable tier history, branch topology, shell-mass ambiguity, scalar-reduction ambiguity, the corrected A4 $q_0 = 0$ endpoint, and positive-time Galerkin scope. It intentionally leaves open: a full three-component derivative Class-II constructive measure; a parameter-identical bridge between the mass anchors; removal of ρ and Class-II regularisers or nonzero η_{shell}; $t = 0$ rates and historical grid error bars; finite-difference Route B; infinite volume and phase transition; and BCC existence, selection, or Sector-B closure.

The highest-value next claim is the full three-component derivative Class-II constructive measure. It must start below T6 and separately establish derivative-current power counting, counterterm classification, uniform stability, tightness, and regulator removal. The present T6 theorem neither prejudges nor bootstraps those gates.

6 Independent reproduction and publication evidence

From either the repository root or the PUBLISHED T6 bundle root, run

```
python codes/foundations/a5_t6_conditional_verify.py
```

The required output is

```
PASS: primary (22/22)
PASS: independent (13/13)
ASSERTS: 35/35
A5-T6-CONDITIONAL-COMPOSITION-INTEGRATED-PASS
Publication: T6-PUBLISHED-OPERATOR-CONFIRMED
```

The additional two checks bind the exact 2026-07-20 confirmation to the v1.0 candidate source, PDF, and commit, and require the T6 reproduction command to be AVAILABLE. The primary audit rehashes all premise records, all six support bundles, the immutable T5 capstone, and the exact current authorities. The independent audit imports no primary implementation and reconstructs the load-bearing topology, Decimal values, and confirmation record directly.

The PUBLISHED T6 bundle is constructed last beside the T5 bundle. Its own manifest inventories all files, reconstructs its content digest, records successful standalone runlogs, and carries the operator-confirmed publication marker. These checks prove record identity and reproducibility, not a new physical prediction.

7 Devil's-advocate review

1. **This is bookkeeping rather than a new physical construction. VALID-with-mitigation.** It is deliberately a conditional-composition theorem. Its value is a checked implication and non-implication firewall.
2. **The T5 inputs violate T6 tier monotonicity. VALID-with-mitigation.** Each T5 hard premise has a separately named hypothesis lift; neither premise is relabelled.
3. **The confirmation could refer to a later reissue. DISMISSED.** The record binds the reviewed v1.0 source and PDF hashes and candidate commit before this v1.1 issue.
4. **Shared geometry implies one parameter set. DISMISSED.** Equation (5.1) independently reconstructs a separation larger than 0.2.
5. **The scalar slice constructs the full Class-II measure. UPHELD against that inference.** Derivative-current products and their renormalization remain a separate open claim.
6. **Perturbative convergence proves the constructive limit. UPHELD against that inference.** Equation (4.2) is a conjunction, not resummation or a dependency arrow.
7. **Positive-time convergence includes $t = 0$. UPHELD against that inference.** The theorem retains $\tau > 0$ and the restarted exact-Galerkin scope.
8. **Hashes prove the component mathematics. VALID-with-mitigation.** Hashes prove exact identity; mathematical force comes from the six confirmed premise packages.
9. **T6 means Sector A is physically complete. UPHELD against that wording.** Full Class-II, infinite volume, BCC, and physical-domain gaps remain explicit.

8 Falsifier and no-overclaim

The theorem is falsified by drift in the immutable T5 capstone, any failed component or support-bundle hash/runlog, a theorem-contract mismatch, any hypothesis count other than seven, an unlifted T5 premise, a broken branch edge, an erased mass fork, a scalar-to-Class-II measure inference, disagreement between audits, or a mismatch in the exact operator-confirmation record.

It does not prove a parameter-identical full-production constructive quantum field theory, a full three-component derivative Class-II Gibbs measure, $\eta_{\text{shell}} \neq 0$, regulator removal, a $t = 0$ rate for merely

H^2 data, historical-grid error bars, finite-difference Route B, unsmeared composite correlations, infinite volume, a phase transition, minimizer uniqueness, BCC existence or selection, Sector-B closure, physical-domain closure, or T7.

Result footer

```
Result ID:                A5-T6-BRANCH-AWARE-CONDITIONAL-COMPOSITION
Precise statement:        Under seven named hypotheses, the full-production
                          implication chain and separate scalar-continuum
                          conjunction hold in their published scopes.
Scope:                    Fixed spectral T3, periods 16, eta_shell=0;
                          branch-aware and non-parameter-identical.
Dependencies:             Six A1-A4 premise claims plus exactly seven named
                          hypotheses, including two T5 premise lifts.
Evidence grade:           ANALYTIC + EXACT + EXECUTED + CONDITIONAL.
Reproduction command:     python codes/foundations/
                          a5_t6_conditional_verify.py
Expected output:          primary 22/22; independent 13/13; ASSERTS 35/35;
                          A5-T6-CONDITIONAL-COMPOSITION-INTEGRATED-PASS.
Falsification gate:       Any premise, hash, hypothesis, edge, fork, audit,
                          exact-confirmation, or standalone-bundle failure.
Tier before / after:      T5 historical capstone / T6 conditional theorem.
No-overclaim statement:   Not a full Class-II constructive measure, parameter
                          identity, infinite volume, phase transition, BCC,
                          Sector-B closure, physical closure, or T7.
Current package status:   T6 operator-confirmed; PUBLISHED bundle built last.
Next research target:     Separate full three-component derivative Class-II
                          constructive-measure claim, starting below T6.
```